	Bronze Layer	Silver Layer	Gold Layer
Definition	**Raw, unprocessed data as-is from sources**	**Clean & standardized data**	**Business-Ready data**
Objective	**Traceability & Debugging**	**(Intermediate Layer) Prepare Data for Analysis**	**Provide data to be consumed for reporting & Analytics**
Object Type	**Tables**	**Tables**	**Views**
Load Method	**Full Load** (Truncate & Insert)	**Full Load** (Truncate & Insert)	**None**
Data Transformation	**None (as-is)**	- Data **Cleaning** - Data **Standardization** - Data **Normalization** - **Derived Columns** - Data **Enrichment**	- Data **Integration** - Data **Aggregation** - **Business Logic & Rules**
Data Modeling	**None (as-is)**	**None (as-is)**	- **Star Schema** - **Aggregated Objects** - **Flat Tables**
Target Audience	- **Data Engineers**	- **Data Analysts** - **Data Engineers**	- **Data Analysts** - **Business Users**



Bronze Layer

Analysing
Interview source systems Experts

Coding
Data **Ingestion**

Validating
Data **Completeness & Schema** Checks

Docs & Version
Documenting
Versioning in **GIT**



Silver Layer

Data Flow
Data Integration

Analysing
Explore & **Understand** the Data

Coding
Data **Cleansing**

Validating
Data **Correctness** checks

Docs & Version
Documenting
Versioning in **GIT**



Gold Layer

Data Model
Data Catalog
Data Flow

Analysing
Explore & **Understand** the Business Objects

Coding
Data **Integration**

Validating
Data **Integration** Checks

Docs & Version
Documenting
Versioning in **GIT**

Source System Interview

Business Context & Ownership

- Who **owns** the **data**?
- What **Business Process** it supports?
- System & Data **documentation**
- Data **Model** & Data **Catalog**

Architecture & Technology Stack

- How is data **stored**?
  (SQL Server, Oracle, AWS, Azure, …)
- What are the **integration capabilities**?
  (API, Kafka, File Extract, Direct DB, ...)

Extract & Load

- **Incremental** vs. **Full Loads** ?
- Data **Scope** & **Historical** Needs
- What is the expected **size** of the **extracts**?
- Are there any data **volume limitations**?
- How to **avoid impacting** the source system's **performance**?
- **authentication** and **authorization**
  (tokens, SSH keys, VPN, IP whitelisting, …)